Renewable and Sustainable Energy Reviews 145 (2021) 111074

Contents lists available at ScienceDirect

Renewable and Sustainable Energy Reviews

journal homepage: http://www.elsevier.com/locate/rser

Recent advancements in nanobubble water technology and its application in energy recovery from organic solid wastes towards a greater environmental friendliness of anaerobic digestion system

Xuezhi Wang [a], Zhongfang Lei [a,*], Kazuya Shimizu [a], Zhenya Zhang [a], Duu-Jong Lee [b]

[a] Graduate School of Life and Environmental Sciences, University of Tsukuba, 1-1-1 Tennodai, Tsukuba, Ibaraki, 305-8572, Japan
[b] Department of Chemical Engineering, National Taiwan University, Taipei, 10617, Taiwan

ARTICLE INFO

Keywords:
Nanobubble water
Bioenergy
Anaerobic digestion
Microaerobic environment
Enzyme activity
Microbial metabolism
Environmental sustainability

ABSTRACT

Anaerobic digestion (AD) is an effective biological process to recover energy from organic solid wastes. There is a growing interest in utilization of renewable or sustainable bioenergy sources as future alternatives to fossil fuels. However, low hydrolysis rate, high heterogeneity of plant cell-wall structure of lignocellulose, process instability at high organic loading rate, and low metabolic activity of microbes limit the practical application of AD of organic solid wastes. A novel nanobubble water (NBW) based-AD system has been proposed to achieve enhanced methane production from typical organic solid wastes. This review summarizes the recent NBW studies, particularly on its utilization in AD processes and its stimulating effects on microorganisms and enzyme activities. The NBW generation approaches and its unique properties are firstly reviewed; then the roles of O_2 in the O_2-containing gas NBW in AD are also outlined and discussed. A deep insight into the feasibility of applying NBW-based AD system in the promotion of energy production from organic solid wastes is provided. Based on the reviewed literature works, the challenges and prospects of adopting the NBW-based AD system as a promising and sustainable technology for energy recovery from organic solid wastes are highlighted.

1. Introduction

Renewable energy is highly demanding due to the shortage of raw materials and the negative impacts of fossil fuels on the environment. The International Energy Agency (IEA) Energy Technology Perspectives 2017 report outlined the requirements for utilization of renewable or sustainable bioenergy, in which biomethane for transport should rise to 10^6 Gigawatt hour (GWh) by 2040 [1]. As reported, in 2015 the biomethane used for vehicles in Germany, Sweden and United States were only 580, 1124 and 2248 GWh, respectively [2]. Anaerobic digestion (AD) is a marketable technology for methane production from organic solid wastes, which has attracted great attention in recent years [3]. In order to meet the increasing energy demand, more developed AD technology for bioenergy conversion from organic wastes is one of the most challenging subjects in the 21st century.

Organic solid wastes generated every day are widely available and can be used in AD to produce energy and other value-added products [4]. In practice, severe inhibition to the decomposition of organic solid wastes with resultant low organic conversion efficiency is usually encountered, mainly attributable to the following aspects: 1) low hydrolysis rate of the organic solid feedstock [5]; 2) the complexity and heterogeneity of plant cell-wall structure of lignocellulosic materials contained in the solid wastes; 3) unstable process at high organic loading rate [6]; 4) low metabolic activity of microbes in AD; 5) natural inhibitors present in the cell wall to resist biological conversion [7]; and 6) toxic and harmful substances contained in the solid wastes [8]. To cope with these problems, various pretreatment techniques have been developed to enhance AD efficiency, including physical (irradiation, extrusion and comminution), physicochemical (hydrothermal and steam explosion), chemical (acid/alkaline, wet explosion, catalyzed hydrothermal and ionic liquids), and biological (bacterial and fungal consortium pretreatment) processes [9]. Besides the above methods, other techniques beyond the pretreatment approaches have been also attempted to improve methane production from organic waste solids for greater environmental friendliness. Most recently, a novel concept of nanobubble water (NBW) based-AD technology was proposed and examined to achieve remarkable performance on methane production from AD of organic solid wastes. It has been recognized that the feasibility of this technology needs to be analyzed systematically in order to

* Corresponding author.
E-mail address: lei.zhongfang.gu@u.tsukuba.ac.jp (Z. Lei).

https://doi.org/10.1016/j.rser.2021.111074
Received 30 September 2020; Received in revised form 29 March 2021; Accepted 1 April 2021
Available online 10 April 2021
1364-0321/© 2021 Elsevier Ltd. All rights reserved.

Abbreviations

AD	Anaerobic digestion
BMP	Biochemical methane potential
COD	Chemical oxygen demand
DHA	Dehydrogenase
DO	Dissolved oxygen
DW	Dry weight
FW	Fresh weight
IEA	International Energy Agency
ISO	International Standards Organization
GWh	Gigawatt hour
LCFA	Long-chain fatty acids
NAD	Nicotinamide adenine dinucleotide
NB	Nanobubble
NBW	Nanobubble water
ORP	Oxidation-reduction potential
ROS	Reactive oxygen species
SDGs	Sustainable development goals
TMP	Thermophilic microaerobic pretreatment
TCA	Tricarboxylic acid
TS	Total solids
VFAs	Volatile fatty acids
VS	Volatile solids

provide comprehensively experimental, technical and theoretical supports for the application of NBW-based AD systems in practice.

NBW has been widely applied in many fields of science and technology such as surfactant-free cleaning and mineral flotation [10,11], biomedicine (enhanced drug susceptibility, cancer cells and intracellular drug delivery) [12–14], and wastewater or groundwater treatment [15]. In the previous works, different terminologies were used for the said bubble water, namely NBW, ultra-fine bubble water, micro-nano bubble water, and so on. In 2017, the International Standards Organization (ISO TC/281) established the standards and regulatory procedures for the fine bubble technology, in which nanobubbles (NBs) are defined as ultrafine bubbles with diameter < 1000 nm (according to ISO 20480–1:2017). Nowadays NBW technology is attracting more interests because of its unique features like non-toxic, low energy consumption and environmentally friendly. NBW has also been proven to have great potentials for enhanced methane production from AD of organic waste solids effectively and economically.

Based on the record in Web of Science (up to 8th March 2021), only 15 publications are available by using keywords "nanobubble water" and "methane". Table 1 mainly shows those works on AD of organic solid wastes. These studies indicate that the NBW-based AD system deserves more in-depth investigations when considering the advantages and potentials of NBW in biological processes. In this regard, few reviews could be found to associate the NBs generation and characterization with the challenges of NBW-based AD systems. In this review, the methods for NBW generation and characteristics of NBs (i.e., stability, specific surface area, mass transfer, and so on) were firstly addressed for its multiple impacts on AD; then the potentials of NBW application to stimulate bioactivities of anaerobic microbes and enzymes were evaluated; and the effects of O_2-containing gas NBW on methane production were analyzed as well. The challenges and future perspectives of NBW-based AD systems were also prospected.

2. Generation of NBW

Johnson and Cooke firstly demonstrated the stable presence of microbubbles formed in seawater, which can maintain stable at least for 22 h [23]. NBW containing 10^6–10^8 gas bubbles with diameter <1 μm can remain as a transparent liquid water [16,17,24,25]. NBs have been suspected for a long time due to the uncertainty of the observed nano-entities being NBs, supramolecular structures or impurities. Most recently, Jadhav and Barigou proved that NBs do exist and are stable in pure water by using various physical and chemical analytical techniques (Fourier transform infrared spectroscopy, Raman spectroscopy, cryo-scanning electron microscope, transmission electron microscope, nanoparticle tracking analysis instrument, and so on) [26]. NBs can be generated by using the following five methods, i.e., cavitation, ceramic membrane filtration, depressurization of saturated solution, gas-water circulation, and electrochemical techniques.

2.1. NBs generation by cavitation

Cavitation can be classified into four categories, namely hydrodynamic, acoustic, optical and particle cavitation. When the liquid passes through a constriction (throttle valve, orifice, venturi, etc.), the kinetic energy/speed of the liquid increases, and the pressure decreases. When the throttling pressure is sufficient to fall below the cavitation threshold pressure (vapor pressure of the medium at the operating temperature), millions of cavities can be generated. This is generally termed as hydraulic cavitation and can be described by Bernoulli's equation.

$$P + \frac{1}{2}\rho U^2 = C$$

where P is the pressure; U is the water flow velocity at a point; ρ is the liquid density; and C is a constant value. When the water flow velocity $U > \sqrt{\frac{2C}{\rho}}$, the pressure would become negative, which is the threshold for starting the cavitation phenomenon [27].

Table 1
Summary of anaerobic digestion enhanced by NBW supplementation.

Feedstock	Type of NBW	Operation temperature (°C)	Increment in methane yield (%)	Increment in substrate removal (%)	Reference
Waste activated sludge	H_2-NBW	36 ± 2	21	–	[16]
Waste activated sludge	N_2-NBW	37 ± 1	29	31 (VFAs production, hydrolysis stage)	[17]
Cellulose	Air-NBW	35 ± 1	18	29 (cellulose content basis)	[18]
Corn straw	O_2-NBW	35 ± 1	22	49 (cellulose and hemicellulose contents basis)	[19]
Cellulose	O_2-NBW	35 ± 1	30	14 (cellulose content basis)	[20]
Food waste	Air-NBW	38 ± 1	24	–	[21]
Lignin	N_2-NBW	37 ± 1	17	10 (lignin content basis)	[22]

NBW, nanobubble water; VFAs, volatile fatty acids.

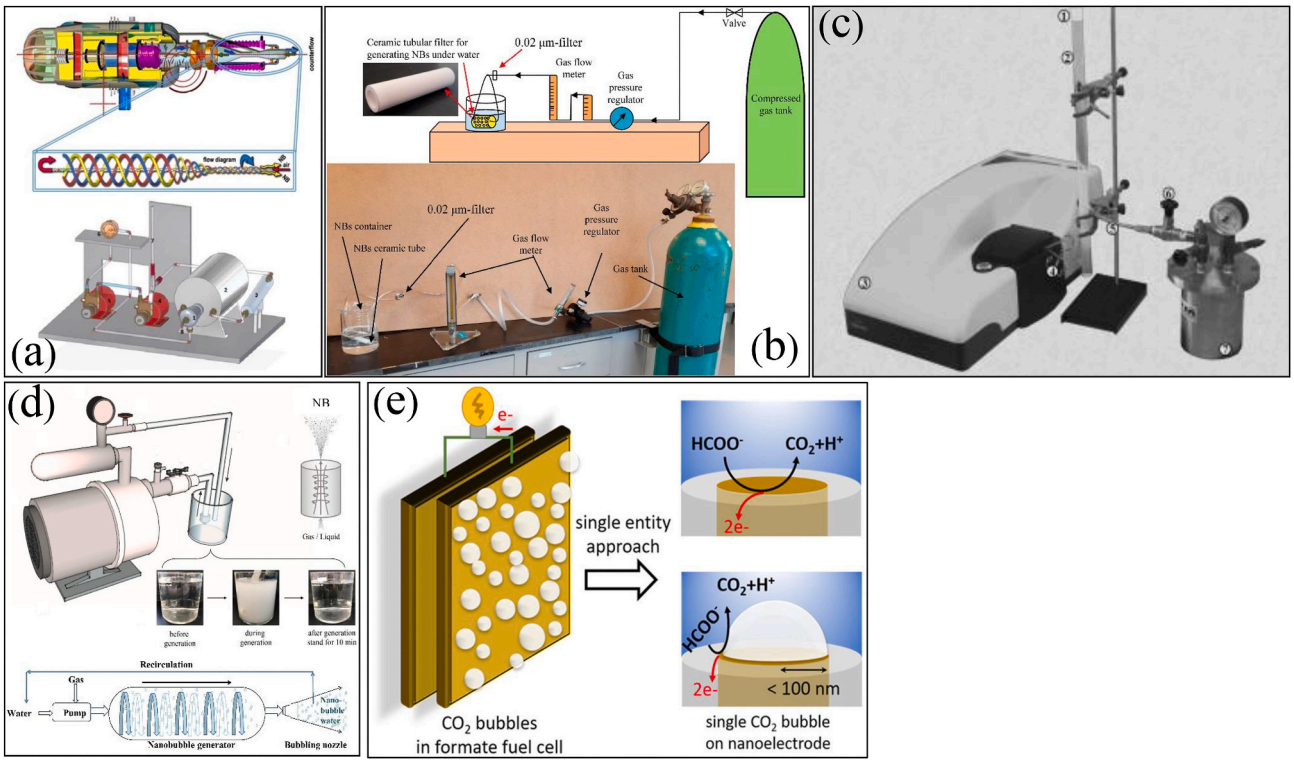


Fig. 1. (a) Schematic of hydrodynamic cavitation to generate NBs [28]; (b) Structure diagram of ceramic membrane filter to produce NBW [32]; (c) Schematic of needle valve to produce NBW [11]; (d) Gas-water recirculation to form NBs [17]; (e) NBs generation by electrochemistry [38].

As shown in Fig. 1a, the Venturi tube is usually utilized for the generation of NBs during hydrodynamic cavitation [28], which can influence the fluid flow by its rough/fractal surface characteristics and transform the system from a liquid/gas mixture to a colloidal phase. During the NBW generation, a critical time is required when the system reaches equilibrium status and the maximum fragmentation in the gas phase occurs, which has been suggested as 30 min by Michailidi et al. [28]. The optimum processing time also depends on the geometrical characteristics of the generator, such as the surface roughness, the counterflow length, the size of the orifices, the effective surface area, the applied feed fluid pressure, and so on [28]. The advantages of this method include low pump power, compact size, capable of generation with high-density NBs of mean diameter below 100 μm, etc. NBW produced by this method has been widely applied in the field of mineral processing and wastewater treatment, due to its energy-efficient and economical [29].

Low pressure can also be created by ultrasonic waves passing the liquid, namely acoustic cavitation. In the NBW generated by acoustic cavitation, the amount of dissolved gas greatly influences the number of nano-entities produced [26], and the number of NBs is reduced in an order of magnitude during acoustic cavitation when compared with other cavitation methods [30]. Optical cavitation can create bubbles by passing a high-intensity light beam (e.g., laser) into the liquid. Particles (such as protons and neutrinos) with high speed that pass through the liquid can also generate cavitation to form bubbles [31]. Obviously, only very few studies are available for the above methods so far due to their high energy consumption and some technical limitations, which is not conducive to practical application.

2.2. NBs generation by ceramic membrane filtration

It has been reported that NBs can be produced by ceramic membranes with uniform nanopores [32]. As shown in Fig. 1b, the NBW generation system is composed of a ceramic tubular filter, a pressurized gas cylinder, a gas flow meter and a gas pressure regulator. This generator can produce NBs with a diameter less than 100 nm when the pore size of the ceramic tube is 100 nm, through which the bubble size and uniformity can be controlled. Some advantages of this method are worth of mentioning: (1) It can produce NBs with strong resistance to corrosive chemicals, extending NBW application to like concentrated acids and high-pressure environments [33]; (2) It can generate NBs independent from liquid circulation, largely reducing the volume, space requirement and operation cost. However, the disadvantage is its longer time, about 90 min continuous injection, for the generated NBs to reach their stabilization than other methods.

2.3. NBs generation by depressurization of saturated solution

Calgaroto et al. noticed that the depressurized air-saturated solution could generate NBs through a needle valve [11]. NBs can be generated simultaneously by gas mass transfer from the liquid to gas phases [34]. By using this method, the air-saturated water solution can be decompressed at a high flow velocity into an empty column (with length of 50 cm and inner diameter of 2 cm) through a needle valve with NBW generated (Fig. 1c). According to the Henry's Law, the amount of gas used for gas nucleus formation depends on the saturation pressure. Thus, the low air/water interfacial tension is also helpful to increase the concentration of NBs (2.5 bar saturation pressure may create 1.6×10^9 NBs/mL), and the high concentration NBs can be maintained in NBW for at least two weeks [35]. High stability and longevity are the advantages of NBs generated by this method, thus, it has been successfully applied in the fields of mineral flotation [11,35] and heavy metals removal [36].

2.4. NBs generation by gas-water circulation

This method combines pressurized mechanical cyclic and spiral liquid flow (gas-water circulation) to generate NBW, which has been applied in the preparation of NBW for the NBW based-AD systems [16–18]. The schematic structure of this type of NBW generator is shown in Fig. 1d. By manually adjusting the liquid in/out flow and gas inflow to

keep the pressure in the chamber at 2.2–2.5 bar, a strong shear force can be created with gas intake speed and deionized water being controlled at 0.05 L/min and 3 L/min, respectively. The NBW generation process generally lasts for 20 min with a consumption of 1 L gas [20]. When some specific gas together with the liquid is injected into the cylinder, normal bubbles can be broken down into NBs due to the created strong shear force. The freshly prepared NBW is like a milky white liquid, owing to the large number of visible microbubbles, which can become very clear and relatively stable after generation for 5–10 min. The NBW generated by this method can achieve good stability of NB size and good performance on methane yield from AD of organic solid wastes [16].

2.5. NBs generation by electrochemical method

Nanoelectrode (4–80 nm) can be used to produce NBs through single-entity electrochemistry. As shown in Fig. 1e, the electrically generated gas can enter the bubble at or near the nanoelectrode/solution/gas three-phase boundary and balance the outflow of the gas at the bubble/solution interface [37]. Formate has been trialed on the Pt nanodisk electrode to successfully form CO_2-NBW by electron oxidation [38]. O_2 and H_2 gases can be produced from the electrolysis of water, then generate stable O_2 NBs and H_2 NBs through bubble nucleation [39]. O_2 NBs can also be produced from H_2O_2 oxidation at positive potential, while H_2 NBs can be generated at negative potential [37]. NBs produced by the electrochemical method are considered more efficient than other methods. However, this method is very complicated with additional contaminants produced (except for O_2 NBs and H_2 NBs), limiting its application in practice when considering its environmental friendliness. In addition, the electric potential needs to be kept sufficient in order to drive the gas generation, greatly increasing its energy consumption.

3. Properties and characteristics of NBW

NBW contains 10^6–10^8 gas bubbles per milliliter with diameter smaller than 1000 nm and possesses unique properties. NBs can attach on the surface of particles or disperse in the liquid phase, which are reported to exist mainly in two forms, i.e., surface and free bubbles. Surface NB refers to a pocket-like space filled with gas (such as air, N_2, O_2, CO_2, etc.), which forms at the interface with solid substrates [40]. Free NB is regarded to exist in the liquid along with Brownian motion [41]. Surface NB has been reported to play a significant role in many modern applications, especially for the separation processes [40]. Table 2 lists the most recent works on NBs and/or NBW applications in other research fields than AD.

Table 2
Most recent works on NBs and/or NBW application in other fields than anaerobic digestion.

Field	Major findings	Reference
Agriculture	○ O_2-NBW irrigation can significantly enhance soil microbial community structure and soil fertility, resulting in improved sugarcane productivity.	[42]
	○ O_2-NBW can improve the mineralization of organic fertilizer with tomato yield increased by 23%.	[43]
	○ Air-, O_2-, N_2- and CO_2– NBW can improve seed germination rates by 6–25%, especially N_2-NBW, due to the generation of exogenous ROS by NBs and efficient nutrient fixation or utilization.	[32]
Water and wastewater treatment	• O_2-NBW can remediate anoxia by increasing DO from 0 to 2.1 mg/L.	[44]
	• O_2-NBW can remediate sediment anoxia and reduce the internal P release in eutrophic lakes.	[45]
	• Air-NBW can promote the development of aerobic biofilm during wastewater treatment.	[46]
	• Water quality can be significantly improved due to 50% increase in COD and NH_4^+ removals.	[47]
	• NBs presence can accelerate the adsorption process of Pb(II) onto activated carbon by 366%, largely decreasing the adsorption equilibrium time and energy consumption.	[48]
Biomedicine	○ O2-NBW has great potential to inhibit tumor growth through decreasing expression of hypoxia-inducible factor.	[13]
Construction and building materials	• NBW can improve the mechanical properties of concrete, including increased compressive strength and tensile strength, and reduced water absorption and chloride ion permeability.	[49]
Membrane cleaning	○ NBW can dramatically clean the membrane by breaking down organic matter due to the produced free radicals from collapse of NBs.	[50]
Flotation	• NBs can increase flotation recovery owing to enhanced attachment of larger bubbles, achieving 95% P_2O_5 recovery within the first 1.5 min.	[51]
Fishery	○ O_2-NBW can extend the survival time of zebrafish in a deoxygenated water system.	[52]

COD, chemical oxygen demand; DO, dissolved oxygen; NBs, nanobubbles; NBW, nanobubble water; ROS, reactive oxygen species.

3.1. Long time stability

The diameter of NBs is less than 5 μm, resulting in the virtual disappearance of buoyant force [53] as well as the physical stability [11]. Air-NBW and O_2-NBW can be stored relatively stable for three months and even longer [28]. Wang et al. also noticed that N_2, CO_2 and H_2 NBs could stably exist in pure water for more than two weeks [16]. As pointed out by Ushikubo et al., the stability of NBs to the electrostatic repulsive force of neighboring bubbles can bring about their long-time independent existence [54]. Chen et al. proposed a high internal gas density model in which the high gas density inside NB could greatly lengthen the lifetime of NB [55]. In addition, when hydrophobic NBs contact with solid surface, the contact angles of 135°–175° can be formed for surface NBs [56]. And the 145° contact angle is regarded as the most stable with the longest lifetime [40]. Ke et al. stated that the negative electrostatic pressure on the interface of NBs can balance the Laplace pressure inside the NBs under alkaline conditions [57]. Thus, NBs are more stable in alkaline solutions, because the more adsorption of OH^- ions at the gas/water interface can enhance the double-layer repulsive force, which then prevents bubble aggregation and coalescence. The stability of NBs also strongly relates to hydrogen bonding at the gas/water interface of NB [58]. The interface of NBs consists of hard hydrogen bonds, similar to hydrogen bonds in ice and natural gas hydrates [59]. As a result, the unique structure and surface properties of NBs make it stable in liquid for a long time.

3.2. High specific surface area

NBs present a high specific surface area, which is beneficial and widely used for flotation in the mineral industry [60]. The surface area increases with the decrease of NB diameter, thus NB reflects higher surface area when compared to microbubbles or general bubbles. To obtain the accurate specific surface area of NBs, use of advanced techniques is necessary. The large specific surface area of NB can result in a considerable capacity for water purification, where is the very suitable space or site for accumulation of microbes and contaminants [61]. NB is able to remain in suspension with contaminants, increasing the probability of pollutant/adsorbent contact [48]. Therefore, biological and chemical reactions can be enhanced on the surface of NB, reflecting a great potential for practical application. NBs can function as carries to transfer the small molecules of organic matters and trace elements, assisting microorganisms in acting on the oligosaccharide layer which

then can be accelerated to break down [18]. Zhang et al. evidenced that NB's high contact angle (low gas-side contact angle) is contributed by the attractive forces between the solid-air and liquid-air interfaces [62]. In addition, Hu and Xia pointed out that the smaller bubbles or their larger specific surface area may exhibit higher mass transfer efficiency [15].

3.3. High mass transfer of NBs

The mass transfer efficiency increases with the decrease in NB radius [63]. The high gas mass transfer efficiency of NBs can dramatically improve the dissolved oxygen (DO) supply for mitigating sediment anoxia and controlling nutrients release from sediments [64]. Xiao and Xu confirmed that O_2-NBW can improve O_2 transfer efficiency by 1.5 times and accelerate biofilm growth [46]. Wang et al. elucidated that O_2 NBs can greatly enhance the photodegradation of oxytetracycline due to its high mass transfer [52]. Phan et al. claimed that high mass transfer of NBs could improve the uniform impregnation efficiency of seasoning boiled egg [65]. The low tendency to coalesce, the large interfacial area, and the low rising velocity can obviously enhance the gas mass transfer efficiency even in a high viscosity liquid. In the field of fermentation, gas transfer greatly influences the whole process, especially on the growth and metabolism of microorganisms [66]. Furthermore, compared with the traditional bubbling aeration process, the oxygen utilization rate and volumetric mass transfer coefficient of O_2-NBW almost doubled in the synthetic wastewater treatment unit, in which the mass transfer rate of O_2 in the O_2-NBW was determined as 125 times faster than that of air macro bubble water [63].

The high viscosity of fermentation broth is usually responsible for its low mass transfer efficiency [66]. As such, the presence of NBs may decrease the viscosity of liquid. However, the effect of viscosity of various gas NBW on mass transfer efficiency during anaerobic fermentation has not been reported. The higher viscosity means more power or operational costs to pump the liquid through the manufacturing equipment, reflecting an enormous economic advantage for the large-scale NBW-based AD systems as the energy consumption for agitation might be minimized. This hypothesis would be further proven by using the developed NBW-based AD systems in the future.

3.4. Negative zeta potential

The zeta potential of NBs in the suspended liquid is considered as a physical property of the boundary layer (also termed as sliding plane) that separates the double layers formed between the counter ions on the bubble interface and the bulk solution [65]. Zeta potential can be used as an important parameter to monitor the NB stability in aqueous solution [67]. Ushikubo et al. proposed that NBs having absolute zeta potential >30 mV are more stable than those <30 mV [54]. The zeta potential of NBs is generally negative over a wide range of pH (3–12) [16]. The negative charge may be associated with the excess of hydroxyl ions relative to hydrogen ions in the first molecular layer of water at the gas-liquid interface [68]. It has also been confirmed that NBs are more stable under alkaline conditions. Intra and intermolecular interactions may occur on the surface of NB via chemical reactions of protonation and deprotonation, such as electrostatic repulsive (for those NBs with the same charge on the surface), electrostatic interactions (for those with opposite charge) and hydrogen bonding interactions (for those with neutral charge) [28]. These interactions also affect the size and the zeta potential value of NBs, which may explain the higher conductivity of NBW and the ability to physically and chemically react on the surface of NBs. Most recently, Wang et al. detected a higher conductivity of NBW than that of deionized water [20].

The negative charge of zeta potential also inhibits the coalescence of NBs in liquid due to the repulsion between the NBs, further contributing to the long lifetime of NBs (over a few months) [28]. The absolute zeta potential depends on the NBW processing time, the kind of gas introduced into the NBW production [28], and the NB movement speed [69]. For example, air (34.0 eV) has a different ionization energy from N_2 (14.5 eV), which may be associated with the higher zeta potential of N_2-NBW than Air-NBW [33,70]. As reported, the ionization potential of O_2 is about 12.1 eV [71], which may result in the higher zeta potential of O_2-NBW than other gas NBWs [28]. Clearly, different gas NBWs may exert different enhancement effects on biomass growth, in which the gas plays a major role. For example, N_2 in N_2-NBW was considered as the governing factor in plant growth, while O_2 in O_2-NBW improved the activity of facultative bacteria [32]. In addition, a relatively high zeta potential has been reported during the collapse of NBs [59]. High zeta potential of NBW would also enhance the adsorption of the positively charged ions.

3.5. Generation of free radicals

Free radicals can be produced from the collapse of ultrasound-induced cavitation bubbles [72]. At the final moment of NBs collapse, the temperature and pressure inside the O_2 NBs can increase up to 2800 K and 4.5 GPa, respectively with resultant NBs wall temperature to 358 K [73]. While this temperature increase in solution is insufficient to thermal dissociation of water molecules and formation of free radicals [74]. Wang et al. stated that free radicals can be generated at the gas-liquid interface from pyrolytic decomposition when NBs collapse occurs [52], which has also been proven by using electron spin resonance spectroscopy technology [75]. Liu et al. experimentally determined the number of •OH radicals generated as the order of 10^{14} per mL in O_2-NBW [76]. The lifetime of free radicals is in the order of 20 ns, which are unstable and difficult to store but can be used for disinfection and sterilization [77]. The production of reactive oxygen species (ROS) by NBs can accelerate the metabolism of living organisms for a long time [76]. In addition, the type of gas inside NBs also affects the quantity of free radicals generated. The NBs containing O_2 were observed to favor the generation of •OH radicals compared to those NBs without O_2 [78]. The free radicals from calcium peroxide are regarded as the contributor to the higher H_2 yield from AD of waste activated sludge [79]. Free radicals can also assist the degradation of various refractory and/or toxic compounds [80], especially for lignin degradation [22].

As it is known, lignin is the most recalcitrant component of lignocellulose that consists of phenylpropane units, which is the main barrier in the use of lignocellulose for bioenergy purpose [81]. Lignin can hinder the direct attack of ligninolytic enzymes to lignin molecules. Yang et al. reported that ROS generated from NBW could facilitate lignin degradation, resulting in 19% increase in lignin degradation rate (64%) during AD when compared to the control [22]. The promotion mechanisms of N_2-NBW on lignin degradation are proposed as follows: firstly, ROS produced from NBs collapse may assist the cleavage of C–C or C–O bonds in lignin molecules, leading to the depolymerization of lignin, and attack the active sites on the benzene ring for opening [22]; secondly, N_2 NB may promote the growth of microorganisms and enhance the oxidative enzyme activity during the AD process [17]. Positive effects of free radicals from NBW on substrates degradation and microbial metabolic activities are also expected in the NBW-based AD systems during the treatment of organic solid wastes.

4. State of the art of NBW-based AD process

4.1. Enhanced microorganisms and microbial communities

Microorganisms are the core in the AD process and they are involved in the biochemical processes of multiple microbial groups. The microbial community is highly sensitive to the surrounding environment and the microbes may adjust the community composition to accommodate the new environment [82]. In the AD system, various anaerobic microorganisms are coexisting in the digester, which can be classified into four categories based on their functions as shown in Fig. 2: hydrolytic,

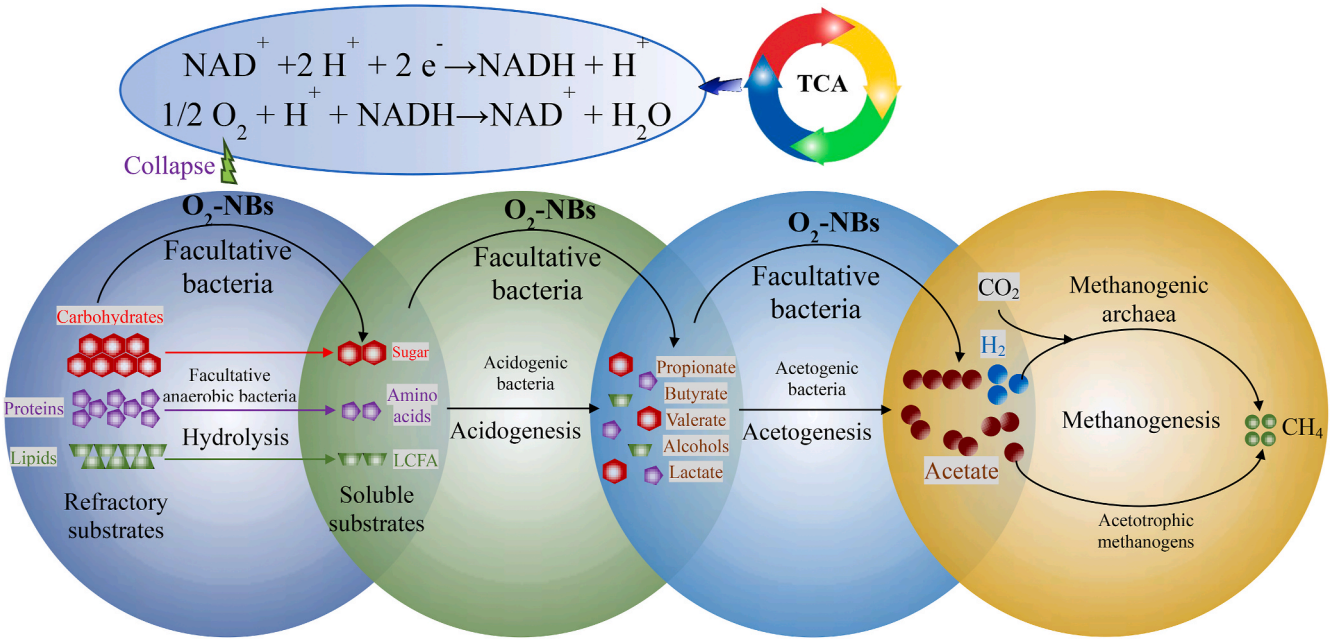


Fig. 2. Syntrophic relationship among the different stages in the O_2-NBW-based AD system. DHA, dehydrogenase; LCFA: long-chain fatty acids; NBs, nanobubbles; TCA, tricarboxylic acid cycle.

acidogenic, acetogenic and methanogenic microorganisms. AD is a multi-stage biochemical process in which the decomposition of complex and refractory organic substrates occurs in series. As reported, the NBW-based AD reactor can be firstly added with the same volume of NBW as that of the inoculum, to which substrates and inoculum are then added according to the feed to microorganism (F/M) ratios [20]. The initial pH is usually adjusted to 7–8 by adding 0.1 M NaOH and/or HCl solution. Prior to methane production, use of N_2 flushing for 2–5 min can create anaerobic environment in the AD reactor which is then operated with temperature-controlled at mesophilic condition (35–38 °C) [17,20].

So far, several research works have investigated the role of NBW in microorganisms. Bhandari et al. observed NBs in the cytoplasm of the cells using the scattering spectral signature with hyperspectral darkfield microscope during 2 h' incubation [83]. This observation adequately proves that the tiny NBs can effectively act on microorganisms. Anaerobic microorganisms (i.e., *Arcobacter* sp., *Azonexus* sp., and *Citrobacter* sp.) can be shifted to aerobic and functional microorganisms (e.g., *Perlucidibaca* sp., *Pseudarcicella* sp., *Rhodoluna* sp., and *Sediminibacterium* sp.) under air NB aeration condition [47]. The changes in structure and dynamics of microbial communities provide a better understanding of the syntrophic interactions in the Air- and CO_2-NBW-based AD process of cellulose [18]: the abundances of hydrolytic bacteria and methanogenic archaea were respectively increased by 52.9% and 63.8% under NBW addition. Similar phenomenon was also noticed in the O_2- or O_2-containing gas NBW-based AD systems of cellulose [20]. Although AD systems of different substrates were operated in different studies, hydrolytic/acidifying bacteria were enriched in all these studies, indicating that NBW supplementation contributes to enhanced hydrolysis and acidification stages, thereby yielding higher volatile fatty acids (VFAs) concentration and then favoring a better methanogenic performance. In addition, Hou et al. [21] found that Air-NBW supplementation could promote not only the hydrolysis/acidification stage but also the methanogenesis stage in the two-stage AD of food waste, reflecting by the enhanced activities of four extracellular hydrolases (alkaline phosphatase, acid phosphatase, α-glucosidase, and protease) and coenzyme F_{420}, respectively.

On the other hand, NBW irrigation can shift the soil bacterial community structure and enhance the potential functions of nitrogen fixation and nitrification, resulting in an increase in plant yield [42]. *Cellvibrio* can degrade refractory materials (such as cellulose) [84], which activity may be stimulated by NBs aeration with enhanced metabolism [85]. NBW supplementation can also enhance methanogenic archaea. Wang et al. found that *Methanosaeta* became the dominant genus under Air-NBW addition [18], which has been recognized as a specialized demulsifying methanogen with a higher affinity for acetate [86,87]. These observations demonstrate that the addition of NBW to AD systems may have a positive effect on microbial community composition for methane production from AD process.

4.2. Stimulated enzyme activities

Degradation of organic wastes is generally catalyzed by various enzymes produced by related microorganisms. Increase in enzyme activity can enhance the AD process. NBW is expected to stimulate various enzymes relating to acidogenesis and methanogenesis. Three mechanisms are involved in the stimulation of enzyme activities under NBW addition. Firstly, the increased mobility of water molecules in NBW can accelerate the mass transfer rate of nutrients from liquid to bacterial cells, which then has an indirect stimulating effect on enzyme activities [18]. Secondly, owing to the hydrophobic surface of NBs, the negative charge on the surface can attract substrates and provide attachment sites for the enzymes, thus more places for enzymatic reactions. Thirdly, increase in enzyme activities might be contributed by the improved metabolic activities of anaerobic microbes [43,76].

As for the whole AD process, the first sub-process is enzymatic hydrolysis, the rate-limiting step in the degradation of refractory substrates (i.e., proteins, polysaccharides and cellulose) to produce biogas [88]. The high mobility NBW was found to assist the cellulase to break down the oligosaccharide layer generated at high cellulose loadings (F/M ratio = 3.5 (VS basis)) [18]. Dehydrogenase (DHA) can be increased by six times under NBs aeration with better removal efficiencies of chemical oxygen demand (COD) and ammonia achieved [46]. DHA is an enzyme responsible for the degradation of organic matter and plays a key role in the microbial metabolism. DHA can also reduce NAD^+ (nicotinamide adenine dinucleotide, as an electron acceptor) and oxidize the substrate. Thus, the enhancement of DHA can strengthen the tricarboxylic acid (TCA) cycle and accelerate the microbial metabolism.

In addition, the introduction of oxygen (collapsed from O_2-NBW) can potentially stimulate the TCA cycle (Fig. 2). These observations to some extent may explain why O_2-containing NB can enhance methane production from AD of lignocellulosic biomass.

NBW supplementation can stimulate more production of extracellular enzymes for enhanced microbial growth and metabolism, which can be indicated by the electron transport system [68]. The higher activity of DHA can efficiently convert substrates to VFAs, beneficial for the subsequent conversion to methane [89]. As recently reported, NBW addition can improve the activities of four extracellular hydrolases (acid phosphatase, alkaline phosphatase, α-glucosidase and protease) by 14–17% during AD of waste activated sludge, achieving greatly improved hydrolysis rate [17]. More specifically, NBW addition can enhance enzyme activity relating to lignocellulose decomposition. Peroxidase is known for its role in depolymerization of lignin [90]. Liu et al. found that NBW could improve peroxidase activity [91]. Wu et al. noticed that O_2-NBW irrigation can significantly increase peroxidase activity in the plantation of tomato with 23% higher tomato yield [43]. Wang et al. claimed that cellulase activity can be elevated by 38% together with more reduction of cellulose crystallinity in the O_2-NBW-based AD system [20], which can enhance cellulose degradation [92]. Thus, NBW addition has a positive effect on the hydrolysis of refractory organic solid wastes and then methane production. The key enzyme, F_{420} is essential and responsible for methanogenesis in the AD process. Liu detected a higher coenzyme F_{420} content in a NBW-based AD system, about 1.23-fold of that in the control [93], which has been also observed in the NBW-based AD of waste activated sludge [94].

Although the above discussion does not cover all the enzymes involved in the AD process, the current observations and knowledge suggest that NBW plays an irreplaceable and important role in the improvement of enzymes activity, most probably due to its unique characteristics like high water molecule mobility, high mass transfer efficiency, no pollution, etc. The NBW-based AD system is expected to significantly improve enzymes activity, which can then greatly enhance methane production and productivity from organic wastes, reflecting its great potentials for achieving sustainable development goals (SDGs).

4.3. Efficient NBW-based AD system with O_2-containing gas NBW supplementation

In general, air or O_2 exposure has been considered to negatively influence the anaerobic microorganisms. However, Angle et al. noticed the occurrence of methanogenesis in the oxygenated freshwater and soil [95]. Table 3 summarizes the previous studies relating to AD process under microaerobic conditions. This kind of microaerobic environment can be easily created by Air-NBW addition with improved hydrolysis/acidification, VFAs production and then methane production from AD of lignocellulosic biomass [18]. Hou et al. also noticed the promoted AD of food waste under the microaerobic environment created by Air-NBW addition [21]. The underlying mechanisms involved are associated with the enriched diversity and enhanced activity of facultative anaerobes that promote the hydrolysis of substrates and then hydrogenotrophic methanogenesis [96]. The addition of O_2-containing gas NBW may realize a delicate balance which not only improves the enzymes activity of microorganisms but also facilitates the metabolism of facultative bacteria. This section provides a detailed information about the enhancement on methane production from AD of organic solid wastes by supplementing O_2-containing gas NBW.

4.3.1. Possibility of microaerobic environment provided by O_2-containing gas NBW

Oxidation-reduction potential (ORP) always changes linearly with the logarithm of O_2 concentration, thus the ORP electrode is sensitive to a small variance in O_2 concentration in the aqueous phase [5]. The standard DO meter or Winkler oxygen titration method is difficult to reflect the oxygen contained in NBs [102]. Nguyen et al. reported that the ORP value of −50 mV corresponds to the detection limit of the DO meter at 0.1 mg/L (3 μM) [96]. Thus, ORP can be used to precisely reflect the oxygen level in AD systems. The ORP value in the range of 0 to −470 mV is defined as microaerobic condition, while the ORP value > 0 mV means aerobic condition [5]. The optimal facultative bacteria can grow and produce AD intermediate products such as VFAs, ethanol and butanol within the ORP range of 0 to −300 mV [5]. O_2-NBW can remediate anoxia by increasing DO from 0 to 2.1 mg/L and ORP by 37% in the overlying water [44]. However, over aeration may exceed the anti-oxidation capacity and O_2 consumption by facultative bacteria, leading to high concentration of ROS that may destroy microbial phospholipid and nucleic acids, and then reduce methane yield [103]. O_2-NBW does not cause over aeration or exceeds the anti-oxidation capacity, as the minimum ORP of pure O_2-NBW is −412 mV in AD process [20], within the ORP range from 0 to −470 mV for an optimal AD process. Thus, the O_2-containing gas NBW may potentially and better provide microaerobic condition for AD systems.

The high stability of O_2-NBW can make O_2 NBs continually be the potent O_2 suppliers in the liquid [104], which could provide O_2 into water slowly and last for a longer time when compared with the ordinary O_2 aeration. Zhang et al. noticed that O_2-NBW continuously improved DO concentrations in the water column, which is essential for

Table 3
Effect of microaerobic environment on biogas and methane yields from AD of lignocellulosic biomass.

Feedstock	Strategy for creating microaerobic environment	Pretreatment conditions	AD scale/operation conditions	Methane yield (L/g-VS)		Reference
				Control	Treatment	
Corn straw	Daily dosing	12.5, 25, 50 or 100 mL/L-reactor/d of air injected into reactor	Lab scale, 200 mL BMP at 55 °C for 30 days	0.186	0.217	[97]
Corn straw	Once	5, 10, 20, 30, or 40 mL/g-$VS_{substrate}$ of oxygen loaded into reactor at 55 °C (TMP)	Lab scale, 500 mL BMP at 37 °C for 60 days	0.28	0.325	[98]
Rice straw	Fungus suspension	*Pleuroyus ostreatus* pretreatment at 28 °C for 20 days	Lab scale, 500 mL BMP at TS 10% and 37 °C for 45 days	0.12	0.263	[99]
Rice straw	Adding supernatant of aerobic sludge and aeration	Rice straw mixed with the supernatant at a ratio of 1:2 (FW basis) was aerated at 0.5 L/min/kg-substrates (DW basis) at 35 °C for 2 days	Lab scale, 1 L BMP at TS 16–28% and 37 °C for 55 days	0.306	0.355	[100]
Primary and waste activated sludge	Aeration (injection of air)	The sludge was aerated at 4 air volume/g TS/min (vmm) for 4 h	Lab scale, 200 mL BMP at 35 °C for 11 days	0.126	0.170	[101]
Napier grass	Intermittent ORP-controlled micro-aeration	Oxygen flow rate controlled at 1 mL/min and ORP of −470 mV during micro-aeration every other day.	Lab scale, 2 L in semi-continuous mode at 37 °C	0.04	0.114	[96]

AD, anaerobic digestion; BMP, biochemical methane potential; DW, dry weight; FW, fresh weight; ORP, oxidation-reduction potential; TMP, thermophilic microaerobic pretreatment; TS, total solids; VS, volatile solids.

the restoration of lakes with algal bloom [45]. Wang et al. observed that O_2-NBW can prolong the survival time of zebrafish and increase the germination rate of *Vallisneria spiralis* seeds in an anaerobic sediment [52]. Xiao and Xu claimed that O_2 NBs can not only continue to dissolve O_2, but also play the role of O_2 storage in the process of dissolution in liquid in situ [46]. Thus, the microaerobic environment created by O_2-NBW may be a promising and environment-friendly technology that would improve the methane yield from AD of refractory materials.

4.3.2. Stability and metabolism of microbes in the microaerobic environment

Refractory organic substrates can be decomposed in AD process through a series of interdependent biochemical steps, where the products of the previous step are served as substrates for the subsequent step (Fig. 2). The O_2 in O_2-NBW can continuously replenish electron acceptors for microorganisms and facilitate the biodegradation and removal of organic pollutants [63]. Hydrolysis of highly recalcitrant feedstocks (energy crops, forest residues and agricultural residues/wastes) can be enhanced by O_2-containing gas NBW, which is critical to dissolve the refractory substrates into soluble substrates for further downstream conversion [105]. The major merit of microaerobic environment is to enhance the activity and diversity of facultative bacteria that could promote hydrolysis [106]. However, when operated at high organic loading rates, VFAs conversion to H_2 is hindered, leading to VFAs accumulation and unstable AD process [107]. Microaerobic environment in combination with aerobic VFAs oxidation by heterotrophs and anaerobic methanogenesis can promote the energetic conversions of intermediates [5]. Under microaerobic conditions, these intermediates can be rapidly consumed by facultative heterotrophs, resulting in a stable AD process. In addition, enhancement of microbial metabolism can also accelerate the consumption of intermediates by facultative heterotrophs.

O_2-NBW can effectively enhance the electron transfer activity of bacteria in AD process due to the high mass transfer of NBs [19,20], which can also increase ATP generation via the TCA cycle (Fig. 2). As noticed, the microbial metabolic functions can be enhanced in tomato growth by NB oxygation (O_2-NBW addition) [43]. Wang et al. obtained 18% increase in methane yield and 20% enhancement in cellulose crystallinity reduction from high loading cellulose AD with the addition of Air-NBW [18], who further proved that O_2-containing gas NBW is able to provide microaerobic environment with resultant enhanced methane production from AD of cellulosic biomass [20]. Air-NBW can improve O_2 content and promote aerobic respiration in the soil [25]. Similarly, O_2 NBs can enhance the activity of aerobic root microorganisms, indirectly promoting plant growth [32]. Through the highly active metabolism, aerobic microbes can grow faster than anaerobic microbes and decompose more organic solid matters into soluble organics that can be utilized during the subsequent steps of AD process including acidification, acetogenesis and methanogenesis.

Besides the role of microaerobic environment, many studies show that NBW can stimulate the metabolism of microorganisms (especially facultative microorganisms), which can be associated with the following aspects. (1) The high specific surface area of NBs [108], the high mobility water molecules in NBW, and the hydrophilicity of NB surface can accelerate the supply of nutrients and trace elements [18]. (2) NBW may alter the abundance of genes. The addition of NBW can reduce the abundance of *hgcA* gene, then reducing the risk of microbial methylmercury production [44]. (3) Hydroxyl radicals and shock waves can be generated at the gas-liquid interface, and thermal decomposition may occur when the NB collapses [109], which could potentially influence the microbial structure and density [110].

5. Challenges and prospects

There has been an increasing interest in the application of NBW to AD systems to enhance the organic waste conversion, methane production, and process stability (Table 1). It is worth noting that NBW supplementation can achieve a higher methane yield from the NBW based-AD system of waste activated sludge or lignocellulosic biomass, increasing by 29% or 30% compared with the control [17,20], which is comparable to or even more efficient than the other enhancement strategies for AD process of organic solid wastes (Table 4). On the other hand, the other enhancement methods for AD of organic solid wastes may bring about environmental issues like possibly secondary pollution, high energy consumption and operation problems such as high operation cost of the AD system in practice. Compared to these enhancement strategies (Table 4), the NBW-based AD technology is more promising, which has been generally considered as a green approach mainly due to its mild reaction condition and non toxic in nature. Restated, NBW supplementation may have the following advantages: (1) O_2-containing gas NBW can create a microaerobic environment favoring the AD process of organic solid wastes; (2) NBW may generate ROS that then promote the hydrolysis of refractory organic solid wastes; and (3) NBW can enhance microbial metabolism and improve enzymes activity in the AD system, and so on. In this context, it is certain that the novel NBW-based AD technology has great potentials for enhanced methane production from AD of organic solid wastes, targeting the sustainable development of the society.

However, attempts are still needed to resolve the major technical obstacle, i.e., increase in AD reactor volume due to NBW addition. Wang et al. first applied NBW pre-augmented anaerobically digested sludge to enhance methane production from AD of corn straw, which can realize high-efficacy AD process of organic solid wastes without increasing the reactor volume due to NBW supplementation [19]. However, further research efforts as follows are necessary to detail the mechanisms and pathways of substrate conversion in the NBW-based AD system in order to realize its high efficiency in industrial and practical applications.

(1) Microbial kinetics, metabolic pathways and genes expression under NBW environment should be explored by using advanced techniques such as stable isotope labeling and metabolomics analysis, which would be helpful to gain a deeper insight into the microbial activities and biochemical pathways of substrate conversion in the NBW-based AD system.
(2) The relationship among the NBW properties in the anaerobic digester (e.g., zeta potential, bubble size, proton spin-spin relaxation time, NBs concentration, specific surface area, and mass transfer of NBs, etc.) is still unclear, which should be further investigated for a better control of the NBW-based AD system.
(3) Regarding the NBW production technology, more focuses should be put on energy consumption reduction and increase in NBs concentration.
(4) A life cycle analysis should be conducted on the specific organic solid waste, the target lignocellulosic biomass in the NBW-based AD system, so as to fully recognize its environmental-friendliness and sustainability.

6. Conclusions

The NBW-based AD system has been reviewed in this paper, with its merits of including enhanced hydrolysis rate of substrates, accelerated VFAs generation and consumption rates, and promoted methane production being outlined. Although recent works confirmed the high efficiency of using NBW-based AD technology to handle the typical organic wastes, this novel AD system has its obstacle to hinder wide applications, especially the concern of increased reactor volume acquired under operation. Nonetheless, the application of NBW-based AD technology as a niche to circular economy in the aspects of organic matter utilization, agricultural waste recycling and renewable energy production, is highlighted. Future works on molecular and microbial analysis, as well as innovative development of cost-effective NBW production technology are proposed to reach industrial application of NBW-

Table 4
Summary of anaerobic digestion of organic solid wastes enhanced by other methods.

Feedstock	Enhancement methods	Anaerobic conditions				Methane yield (mL/g-VS)	Increment in methane yield (%)	Increment in substrate removal (%)	Reference
		Operation temperature (°C)	Fermentation duration (d)	Initial pH	Digester type				
Corn straw	Alkaline/micro-aerobic pretreatment	37 (mesophilic)	43	–	Batch	242	11	–	[111]
Corn Straw	Micro-aerobic pretreatment	37 (mesophilic)	50	7.4	Batch	271	17	14 (crystallinity degree basis)	[112]
WAS	Oxidative pretreatment	35 (mesophilic)	40	–	Batch	204	24	–	[113]
Food waste	Zero-valent iron addition	35 (mesophilic)	31	–	Two-stage, batch	366	13	–	[114]
Food waste	Biochar addition	35 (mesophilic)	60	–	Batch	494	7	–	[115]
Food waste	Trance element addition	37 (mesophilic)	30	–	Batch	550	22	–	[116]
WAS	Freezing sludge with nitrite addition	35 (mesophilic)	36	7.0	Batch	233	30	–	[117]
WAS and wheat straw	Hydrothermal pretreatment	37 (mesophilic)	40	–	Batch	226	19	16 (VS basis)	[118]
Agricultural wastes	Oxidative pretreatment	35 (mesophilic)	96	–	Batch	301	34	25 (VS basis)	[119]
Corn straw	Enzymatic pretreatment	37 (mesophilic)	30	–	Batch	278	28	28 (OM basis)	[120]

OM, organic matter; WAS, waste activated sludge; VS, volatile solid.

based AD systems.

CRediT author contribution statement

Xuezhi Wang: Writing-original draft, Conceptualization. Zhongfang Lei: Funding acquisition, Writing-review & editing, Supervision. Kazuya Shimizu: Writing - review & editing. Zhenya Zhang: Writing - review & editing. Duu-Jong Lee: Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by JSPS KAKENHI Grant Numbers JP18H03403 and JP16H06382. Mr. Xuezhi Wang would like to thank the financial support by the China Scholarship Council (No. 201706400077) for his PhD study at University of Tsukuba, Japan.

References

[1] International Energy Agency (IEA). Energy technology perspectives. Available at: http://www.iea.org/etp2017/summary/. [Accessed 3 March 2021].
[2] Svensson M. Challenges and opportunities for the future biomethane powered automotive market. Gent: EBA Conference; 2016. Available at: http://www.european-biogas.eu. [Accessed 3 March 2021].
[3] Masebinua SO, Akinlabia ET, Muzendab E, Aboyadec AO. A review of biochar properties and their roles in mitigating challenges with anaerobic digestion. Renew Sustain Energy Rev 2019;103:291–307.
[4] Brémond U, de Buyer R, Steyer J-P, Bernet N, Carrere H. Biological pretreatments of biomass for improving biogas production: an overview from lab scale to full-scale. Renew Sustain Energy Rev 2018;90:583–604.
[5] Nguyen D, Khanal SK. A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process. Biotechnol Adv 2018;36:1971–83.
[6] Mlinar S, Weig AR, Freitag R. Influence of mixing and sludge volume on stability, reproducibility, and productivity of laboratory-scale anaerobic digestion. Bioresour Technol Rep 2020;11:100444.
[7] Onumaegbu C, Mooney J, Alaswad A, Olabi AG. Pre-treatment methods for production of biofuel from microalgae biomass. Renew Sustain Energy Rev 2018;93:16–26.
[8] Zhen G, Lu X, Li Y-Y, Zhao Y. Combined electrical-alkali pretreatment to increase the anaerobic hydrolysis rate of waste activated sludge during anaerobic digestion. Appl Energ 2014;128:93–102.
[9] Zheng Y, Zhao J, Xu F, Li Y. Pretreatment of lignocellulosic biomass for enhanced biogas production. Prog Energ Combust 2014;42:35–53.
[10] Zhu J, An H, Alheshibri M, Liu L, Terpstra PMJ, Liu G, Craig VSJ. Cleaning with bulk nanobubbles. Langmuir 2016;32:11203–11.
[11] Calgaroto S, Wilberg KQ, Rubio J. On the nanobubbles interfacial properties and future applications in flotation. Miner Eng 2014;60:33–40.
[12] Suzuki M, Koshiyama K, Shinohara F, Mori S, Ono M, Tomita Y, Yano T, Fujikawa S, Vassaux G, Kodama T. Nanobubbles enhanced drug susceptibility of cancer cells using ultrasound. Int Congr Ser 2005;1284:338–9.
[13] Mahjour A, Khazaei M, Nourmohammadi E, Khoshdel-Sarkarizi H, Ebrahimzadeh-Bideskan A, Rahimi HR, Afshar AS. Evaluation of antitumor effect of oxygen nanobubble water on breast cancer-bearing BALB/c mice. J Cell Biochem 2019;120:15546–52.
[14] Xie X, Lin W, Liu H, Deng J, Chen Y, Liu H, Fu X, Yang Y. Ultrasound-responsive nanobubbles contained with peptide–camptothecin conjugates for targeted drug delivery. Drug Deliv 2016;23:2756–64.
[15] Hu L, Xia Z. Application of ozone micro-nano-bubbles to groundwater remediation. J Hazard Mater 2018;342:446–53.
[16] Wang D, Yang X, Tian C, Lei Z, Kobayashi N, Kobayashi M, Adachi Y, Shimizu K, Zhang Z. Characteristics of ultra-fine bubble water and its trials on enhanced methane production from waste activated sludge. Bioresour Technol 2019;273:63–9.
[17] Yang X, Nie J, Wang D, Zhao Z, Kobayashi M, Adachi Y, Shimizu K, Lei Z, Zhang Z. Enhanced hydrolysis of waste activated sludge for methane production via anaerobic digestion under N_2-nanobubble water addition. Sci Total Environ 2019;693:133524.
[18] Wang X, Yuan T, Guo Z, Han H, Lei Z, Shimizu K, Zhang Z, Lee D-J. Enhanced hydrolysis and acidification of cellulose at high loading for methane production via anaerobic digestion supplemented with high mobility nanobubble water. Bioresour Technol 2020;297:122499.
[19] Wang X, Lei Z, Shimizu K, Zhang Z, Lee D-J. Improved methane production from corn straw using anaerobically digested sludge pre-augmented by nanobubble water. Bioresour Technol 2020;311:123479.
[20] Wang X, Yuan T, Lei Z, Kobayashi M, Adachi Y, Shimizu K, Lee D-J, Zhang Z. Supplementation of O_2-containing gas nanobubble water to enhance methane production from anaerobic digestion of cellulose. Chem Eng J 2020;398:125652.
[21] Hou T, Zhao J, Lei Z, Shimizu K, Zhang Z. Enhanced energy recovery via separate hydrogen and methane production from two-stage anaerobic digestion of food waste with nanobubble water supplementation. Sci Total Environ 2021;761:143234.
[22] Yang X, Nie J, Wei Y, Zhao Z, Shimizu K, Lei Z, Zhang Z. Simultaneous enhancement on lignin degradation and methane production from anaerobic co-digestion of waste activated sludge and alkaline lignin supplemented with N_2-nanobubble water. Bioresour Technol Rep 2020;11:100470.
[23] Johnson BD, Cooke RC. Generation of stabilized microbubbles in seawater. Science 1981;213:209–11.
[24] Liu S, Kawagoe Y, Makino Y, Oshita S. Effects of nanobubbles on the physicochemical properties of water: the basis for peculiar properties of water containing nanobubbles. Chem Eng Sci 2013;93:250–6.

[25] Liu Y, Zhou Y, Wang T, Pan J, Zhou B, Muhammad T, Zhou C, Li Y. Micro-nano bubble water oxygation: synergistically improving irrigation water use efficiency, crop yield and quality. J Clean Prod 2019;222:835–43.
[26] Jadhav AJ, Barigou M. Bulk nanobubbles or not nanobubbles: that is the question. Langmuir 2020;36:1699–708.
[27] Zhou ZA, Xu Z, Finch JA, Masliyah JH, Chow RS. On the role of cavitation in particle collection in flotation - A critical review. II. Miner Eng 2009;22:419–33.
[28] Michailidi ED, Bomis G, Varoutoglou A, Kyzas GZ, Mitrikas G, Mitropoulos AC, Efthimiadou EK, Favvas EP. Bulk nanobubbles: production and investigation of their formation/stability mechanism. J Colloid Interface Sci 2020;564:371–80.
[29] Azevedo A, Oliveira H, Rubio J. Bulk nanobubbles in the mineral and environmental areas: updating for research and applications. Adv Colloid Interfac 2019;271. UNSP 101992.
[30] Tomita Y, Shima A. High-speed photographic observations of laser-induced cavitation bubbles in water. Acustica 1990;71:161–71.
[31] Manickam S, Ashokkumar M. Cavitation: a novel energy-efficient technique for the generation of nanomaterials. CRC Press; 2014.
[32] Ahmed AKA, Shi X, Hua L, Manzueta L, Qing W, Marhaba T, Zhang W. Influences of air, oxygen, nitrogen, and carbon dioxide nanobubbles on seed germination and plant growth. J Agric Food Chem 2018;66:5117–24.
[33] Ahmed AKA, Sun C, Hua L, Zhang Z, Zhang Y, Zhang W, Marhaba T. Generation of nanobubbles by ceramic membrane filters: the dependence of bubble size and zeta potential on surface coating, pore size and injected gas pressure. Chemosphere 2018;203:327–35.
[34] Calgarato S, Azevedo A, Rubio J. Separation of amine-insoluble species by flotation with nano and microbubbles. Miner Eng 2016;89:24–9.
[35] Azevedo A, Etchepare R, Calgaroto S, Rubio J. Aqueous dispersions of nanobubbles: generation, properties and features. Miner Eng 2016;94:29–37.
[36] Silva RDR, Rodrigues RT, Azevedo AC, Rubio J. Calcium and magnesium ion removal from water feeding a steam generator by chemical precipitation and flotation with micro and nanobubbles. Environ Technol 2020;41:2210–8.
[37] Ren H, German SR, Edwards MA, Chen Q, White HS. Electrochemical generation of individual O_2 nanobubbles via H_2O_2 oxidation. J Phys Chem Lett 2017;8:2450–4.
[38] Ren H, Edwards MA, Wang Y, White HS. Electrochemically controlled nucleation of single CO_2 nanobubbles via formate oxidation at Pt nanoelectrodes. J Phys Chem Lett 2020;11:1291–6.
[39] German SR, Edwards MA, Chen Q, Liu Y, Luo L, White HS. Electrochemistry of single nanobubbles. Estimating the critical size of bubble-forming nuclei for gas-evolving electrode reactions. Faraday Discuss 2016;193:223–40.
[40] Theodorakis PE, Che Z. Surface nanobubbles: theory, simulation, and experiment. A review. Adv Colloid Interfac 2019;272. UNSP 101995.
[41] Kim M, Song S, Kim W, Han J-G. An experimental study on bubble collapsing effect of nanobubble using ultrasonic wave. J Nanosci Nanotechnol 2020;20:636–42.
[42] Zhou Y, Bastida F, Zhou B, Sun Y, Gu T, Li S, Li Y. Soil fertility and crop production are fostered by micro-nano bubble irrigation with associated changes in soil bacterial community. Soil Biol Biochem 2020;141:107663.
[43] Wu Y, Lyu T, Yue B, Tonoli E, Verderio EAM, Ma Y, Pan G. Enhancement of tomato plant growth and productivity in organic farming by agri-nanotechnology using nanobubble oxygation. J Agric Food Chem 2019;67:10823–31.
[44] Ji X, Liu C, Pan G. Interfacial oxygen nanobubbles reduce methylmercury production ability of sediments in eutrophic waters. Ecotoxicol Environ Saf 2020;188:109888.
[45] Zhang H, Chen J, Han M, An W, Yu J. Anoxia remediation and internal loading modulation in eutrophic lakes using geoengineering method based on oxygen nanobubbles. Sci Total Environ 2020;714:136766.
[46] Xiao W, Xu G. Mass transfer of nanobubble aeration and its effect on biofilm growth: microbial activity and structural properties. Sci Total Environ 2020;703:134976.
[47] Wu Y, Lin H, Yin W, Shao S, Lv S, Hu Y. Water quality and microbial community changes in an urban river after micro-nano bubble technology in situ treatment. Water 2019;11:66.
[48] Kyzas GZ, Bomis G, Kosheleva RI, Efthimiadou EK, Favvasa EP, Kostoglou M, Mitropoulos AC. Nanobubbles effect on heavy metal ions adsorption by activated carbon. Chem Eng J 2019;356:91–7.
[49] Khoshroo M, Javid AAS, Katebi A. Effects of micro-nano bubble water and binary mineral admixtures on the mechanical and durability properties of concrete. Construct Build Mater 2018;164:371–85.
[50] Ghadimkhani A, Zhang W, Marhaba T. Ceramic membrane defouling (cleaning) by air nano bubbles. Chemosphere 2016;146:379–84.
[51] Rosa AF, Rubio J. On the role of nanobubbles in particle-bubble adhesion for the flotation of quartz and apatitic minerals. Miner Eng 2018;127:178–84.
[52] Wang L, Ali J, Wang Z, Oladoja NA, Cheng R, Zhang C, Mailhot G, Pan G. Oxygen nanobubbles enhanced photodegradation of oxytetracycline under visible light: synergistic effect and mechanism. Chem Eng J 2020;388:124227.
[53] Zimmerman WB, Tesař V, Bandulasena HCH. Towards energy efficient nanobubble generation with fluidic oscillation. Curr Opin Colloid In 2011;16:350–6.
[54] Ushikubo FY, Furukawa T, Nakagawa R, Enari M, Makino Y, Kawagoe Y, Shiina T, Oshita S. Evidence of the existence and the stability of nano-bubbles in water. Colloids Surf, A 2010;361:31–7.
[55] Chen C, Li J, Zhang X. The existence and stability of bulk nanobubbles: a long-standing dispute on the experimentally observed mesoscopic inhomogeneities in aqueous solutions. Commun Theor Phys 2020;72:037601. UNSP.
[56] Borkent BM, de Beer S, Mugele F, Lohse D. On the shape of surface nanobubbles. Langmuir 2010;26:260–8.
[57] Ke S, Xiao W, Quan N, Dong Y, Zhang L, Hu J. Formation and stability of bulk nanobubbles in different solutions. Langmuir 2019;35:5250–6.
[58] Ohgaki K, Khanh NQ, Joden Y, Tsuji A, Nakagawa T. Physicochemical approach to nanobubble solutions. Chem Eng Sci 2010;65:1296–300.
[59] Agarwal A, Ng WJ, Liu Y. Principle and applications of microbubble and nanobubble technology for water treatment. Chemosphere 2011;84:1175–80.
[60] Michailidi ED, Bomis G, Varoutoglou A, Efthimiadou EK, Mitropoulos AC, Favvas EP. Chapter 4-Fundamentals and applications of nanobubbles. Interface Sci Technol 2019;30:69–99.
[61] Yoshida A, Takahashi O, Ishii Y, Sekimoto Y, Kurata Y. Water purification using the adsorption characteristics of microbubbles. Jpn J Appl Phys 2008;47:6574–7.
[62] Zhang XH, Quinn A, Ducker WA. Nanobubbles at the interface between water and a hydrophobic solid. Langmuir 2008;24:4756–64.
[63] Li H, Hu L, Song D, Lin F. Characteristics of micro-nano bubbles and potential application in groundwater bioremediation. Water Environ Res 2014;86:844–51.
[64] Zhang H, Lyu T, Bi L, Tempero G, Hamilton DP, Pan G. Combating hypoxia/anoxia at sediment-water interfaces: a preliminary study of oxygen nanobubble modified clay materials. Sci Total Environ 2018;637:550–60.
[65] Phan KKT, Truong T, Wang Y, Bhandari B. Nanobubbles: fundamental characteristics and applications in food processing. Trends Food Sci Technol 2020;95:118–30.
[66] Weber J, Agblevor FA. Microbubble fermentation of *Trichoderma reesei* for cellulase production. Process Biochem 2005;40:669–76.
[67] Parmar R, Majumder SK. Microbubble generation and microbubble-aided transport process intensification-a state-of-the-art report. Chem Eng Process 2013;64:79–97.
[68] Oh SH, Kim J-M. Generation and stability of bulk nanobubbles. Langmuir 2017;33:3818–23.
[69] Takahashi M. ζ potential of microbubbles in aqueous solutions: electrical properties of the gas-water interface. J Phys Chem B 2005;109:21858–64.
[70] Ahmed AKA, Sun C, Hua L, Zhang Z, Zhang Y, Marhaba T, Zhang W. Colloidal properties of air, oxygen, and nitrogen nanobubbles in water: effects of ionic strength, natural organic matters, and surfactants. Environ Eng Sci 2016;35:720–7.
[71] Samson JAR, Cairns RB. Ionization potential of O_2. J Opt Soc Am 1966;56:769–75.
[72] Li P, Takahashi M, Chiba K. Enhanced free-radical generation by shrinking microbubbles using a copper catalyst. Chemosphere 2009;77:1157–60.
[73] Yasui K, Tuziuti T, Kanematsu W. Extreme conditions in a dissolving air nanobubble. Phys Rev E 2016;94:013106.
[74] Yasui K, Tuziuti T, Kanematsu W. High temperature and pressure inside a dissolving oxygen nanobubble. Ultrason Sonochem 2019;55:308–12.
[75] Takahashi M, Chiba K, Li P. Free-radical generation from collapsing microbubbles in the absence of a dynamic stimulus. J Phys Chem B 2007;111:1343–7.
[76] Liu S, Oshita S, Kawabata S, Makino Y, Yoshimoto T. Identification of ROS produced by nanobubbles and their positive and negative effects on vegetable seed germination. Langmuir 2016;32:11295–302.
[77] Yasui K, Tuziuti T, Kanematsu W. Mysteries of bulk nanobubbles (ultrafine bubbles); stability and radical formation. Ultrason Sonochem 2018;48:259–66.
[78] Li P, Takahashi M, Chiba K. Degradation of phenol by the collapse of microbubbles. Chemosphere 2009;75:1371–5.
[79] Wang D, Zhang D, Xu Q, Liu Y, Wang Q, Ni B-J, Yang Q, Li X, Yang F. Calcium peroxide promotes hydrogen production from dark fermentation of waste activated sludge. Chem Eng J 2019;355:22–32.
[80] Khan JA, He X, Shah NS, Khan HM, Hapeshi E, Fatta-Kassinos D, Dionysiou DD. Kinetic and mechanism investigation on the photochemical degradation of atrazine with activated H_2O_2, $S_2O_8^{2-}$ and HSO_5^-. Chem Eng J 2014;252:393–403.
[81] Mirmohamadsadeghi S, Karimi K, Azarbaijanic R, Yeganeh LP, Angelidaki I, Nizami A-S, Bhat R, Dashora K, Vijay VK, Aghbashlo M, Gupta VK, Tabatabaei M. Pretreatment of lignocelluloses for enhanced biogas production: a review on influencing mechanisms and the importance of microbial diversity. Renew Sustain Energy Rev 2021;135:110173.
[82] Xie Y, Wang J, Wu Y, Ren C, Song C, Yang J, Yu H, Giesy JP, Zhang X. Using in situ bacterial communities to monitor contaminants in river sediments. Environ Pollut 2016;212:348–57.
[83] Bhandari P, Wang X, Irudayaraj J. Oxygen nanobubble tracking by light scattering in single cells and tissues. ACS Nano 2017;11:2682–8.
[84] Berg B, Hofsten BV, Pettersson G. Electron microscopie observations on the degradation of cellulose fibres by *Cellvibrio fulvus* and *Sporocytophaga myxococcoides*. J Appl Microbiol 1972;35:215–9.
[85] Sun Y, Wang S, Niu J. Microbial community evolution of black and stinking rivers during in situ remediation through micro-nano bubble and submerged resin floating bed technology. Bioresour Technol 2018;258:187–94.
[86] Garcia J-L, Patel BKC, Ollivier B. Taxonomic, phylogenetic, and ecological diversity of methanogenic archaea. Anaerobe 2000;6:205–26.
[87] Conklin A, Stensel HD, Ferguson J. The growth kinetics and competition between *methanosarcina* and *methanosaeta* in mesophilic anaerobic digestion. Water Environ Res 2006;78:486–96.
[88] Costa THF, Eijsink VGH, Horn SJ. The use of lytic polysaccharide monooxygenases in anaerobic digestion of lignocellulosic materials. Biotechnol Biofuels 2019;12:270.
[89] Feng H, Hu L, Mahmood Q, Fang C, Qiu C, Shen D. Effects of temperature and feed strength on a carrier anaerobic baffled reactor treating dilute wastewater. Desalination 2009;239:111–21.

[90] Sinsabaugh RL. Phenol oxidase, peroxidase and organic matter dynamics of soil. Soil Biol Biochem 2010;42:391–404.
[91] Liu S, Oshita S, Kawabata S, Thuyet DQ. Nanobubble water's promotion effect of barley (*Hordeum vulgare L.*) sprouts supported by RNA-seq analysis. Langmuir 2017;33:12478–86.
[92] Oliveira DM, Hoshino ÉP, Mota TR, Marchiosi R, Ferrarese-Filho O, dos Santos WD. Modulation of cellulase activity by lignin-related compounds. Bioresour Technol Rep 2020;10:100390.
[93] Liu Y. Application of micro-nano bubbles on bioconversion of carbon dioxide and hydrogen to methane. Doctoral Dissertation, University of Tsukuba; 2018.
[94] Wang D. Characteristics of ultra-fine bubble water and its effects on anaerobic digestion of waste activated sludge. Doctoral Dissertation, University of Tsukuba; 2019.
[95] Angle JC, Morin TH, Solden LM, Narrowe AB, Smith GJ, Borton MA, Rey-Sanchez C, Daly RA, Mirfenderesgi G, Hoyt DW, Riley WJ, Miller CS, Bohrer G, Wrighton KC. Methanogenesis in oxygenated soils is a substantial fraction of wetland methane emissions. Nat Commun 2017;8:1567.
[96] Nguyen D, Wu Z, Shrestha S, Lee P-H, Raskin L, Khanal SK. Intermittent micro-aeration: new strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion. Water Res 2019;166:115080.
[97] Fu S-F, Wang F, Shi X-S, Guo R-B. Impacts of microaeration on the anaerobic digestion of corn straw and the microbial community structure. Chem Eng J 2016; 287:523–8.
[98] Fu S-F, Wang F, Yuan X-Z, Yang Z-M, Luo S-J, Wang C-S, Guo R-B. The thermophilic (55°C) microaerobic pretreatment of corn straw for anaerobic digestion. Bioresour Technol 2015;175:203–8.
[99] Mustafa AM, Poulsen TG, Sheng K. Fungal pretreatment of rice straw with Pleurotus ostreatus and Trichoderma reesei to enhance methane production under solid-state anaerobic digestion. Appl Energ 2016;180:661–71.
[100] Zhou Y, Li C, Nges IA, Liu J. The effects of pre-aeration and inoculation on solid-state anaerobic digestion of rice straw. Bioresour Technol 2017;224:78–86.
[101] Ruan D, Zhou Z, Pang H, Yao J, Chen G, Qiu Z. Enhancing methane production of anaerobic sludge digestion by microaeration: enzyme activity stimulation, semi-continuous reactor validation and microbial community analysis. Bioresour Technol 2019;289:121643.
[102] Alheshibri M, Qian J, Jehannin M, Craig VSJ. A history of nanobubbles. Langmuir 2016;32:11086–100.
[103] Xu S, Selvam A, Wong JWC. Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste. Waste Manag 2014;34:363–9.
[104] Ebina K, Shi K, Hirao M, Hashimoto J, Kawato Y, Kaneshiro S, Morimoto T, Koizumi K, Yoshikawa H. Oxygen and air nanobubble water solution promote the growth of plants, fishes, and mice. PloS One 2013;8:e65339.
[105] Shrestha S, Fonoll X, Khanal SK, Raskin L. Biological strategies for enhanced hydrolysis of lignocellulosic biomass during anaerobic digestion: current status and future perspectives. Bioresour Technol 2017;245:1245–57.
[106] Zhu M, Lü F, Hao L-P, He P-J, Shao L-M. Regulating the hydrolysis of organic wastes by micro-aeration and effluent recirculation. Waste Manag 2009;29: 2042–50.
[107] Leng L, Yang P, Singh S, Zhuang H, Xu L, Chen W-H, Dolfing J, Li D, Zhang Y, Zeng H, Chu W, Lee P-H. A review on the bioenergetics of anaerobic microbial metabolism close to the thermodynamic limits and its implications for digestion applications. Bioresour Technol 2017;247:1095–106.
[108] Temesgen T, Bui TT, Han M, Kim T-i, Park H. Micro and nanobubble technologies as a new horizon for water-treatment techniques: a review. Adv Colloid Interfac 2017;246l:40–51.
[109] Sivasankar T, Moholkar VS. Physical insights into the sonochemical degradation of recalcitrant organic pollutants with cavitation bubble dynamics. Ultrason Sonochem 2009;16:769–81.
[110] Sumikura M, Hidaka M, Murakami H, Nobutomo Y, Murakami T. Ozone micro-bubble disinfection method for wastewater reuse system. Water Sci Technol 2007; 56:53–61.
[111] Fu S-F, Chen K-Q, Sun W-X, Zhu R, Zheng Y, Zou H. Improved methane production of corn straw by the stimulation of calcium peroxide. Energy Convers Manag 2018;164:36–41.
[112] Xu W, Fu S, Yang Z, Lu J, Guo R. Improved methane production from corn straw by microaerobic pretreatment with a pure bacteria system. Bioresour Technol 2018;259:18–23.
[113] Hu J, Guo B, Li Z, Eshtiaghi N, Tao W. Revealing the mechanisms for potassium ferrate affecting methane production from anaerobic digestion of waste activated sludge. Bioresour Technol 2020;317:124022.
[114] Yuan T, Bian S, Ko JH, Liu J, Shi X, Xu Q. Exploring the roles of zero-valent iron in two-stage food waste anaerobic digestion. Waste Manag 2020;107:91–100.
[115] Yuan T, Shi X, Sun R, Ko JH, Xu Q. Simultaneous addition of biochar and zero-valent iron to improve food waste anaerobic digestion. J Clean Prod 2021;278: 123627.
[116] Shamurad B, Sallis P, Petropoulos E, Tabraiz S, Ospina C, Leary P, et al. Stable biogas production from single-stage anaerobic digestion of food waste. Appl Energ 2020;263:114609.
[117] Liu X, Huang X, Wu Y, Xu Q, Du M, Wang D, Yang Q, Liu Y, Ni B-J, Yang G, Yang F, Wang Q. Activation of nitrite by freezing process for anaerobic digestion enhancement of waste activated sludge: performance and mechanisms. Chem Eng J 2020;387:124147.
[118] Tian W, Chen Y, Shen Y, Zhong C, Gao M, Shi D, He Q, Gu L. Effects of hydrothermal pretreatment on the mono- and co-digestion of waste activated sludge and wheat straw. Sci Total Environ 2020;732:139312.
[119] Almomani F, Bhosale RR, Khraisheh MAM, Shawaqfah M. Enhancement of biogas production from agricultural wastes via pre-treatment with advanced oxidation processes. Fuel 2019;253:964–74.
[120] Wang X, Cheng S, Li Z, Men Y, Wu J. Impacts of cellulase and amylase on enzymatic hydrolysis and methane production in the anaerobic digestion of corn straw. Sustainability 2020;12:5453.